

Expanding versus greening? Long-term energy and emission transitions in Mozambique

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ABSTRACT

Mozambique is rapidly emerging as one of the leading energy producers in Africa. We develop a comprehensive scenario analysis to review the energy revolution in Mozambique. Like much of Africa, Mozambique features a competition between development of its vast reserves of renewable energy and fossil fuels. We assess the implied energy and emissions transitions as well as the potential impact of a range of sustainable policy options. Our analysis reveals an emerging ‘energy dichotomy’, where a spectacular expansion of energy production goes together with an only gradual energy consumption transition. We find that over time the share of modern renewable energy sources tends to marginalize in the energy production mix while remaining stable in the energy consumption mix. About one-third of the GHG emissions associated with energy production can be linked to energy export. Also, we find that sustainable energy policies at the supply side lead to far higher cumulative emission reductions than demand-side policies. Finally, we show that even in the long run, a complete ban on thermal electricity generation capacity in Mozambique would by no means lead to shortages in the domestic electricity market but only slightly limit potential electricity export.

1. Introduction

In its latest Africa Energy Outlook, the International Energy Agency (IEA, 2014) observes that Africa's energy sector is vital to the continent's future development and yet remains the most relatively poorly understood part of the global energy system. Fortunately, Africa is rich in renewable energy sources: its energy mix is dominated by bioenergy, it has a huge solar energy potential, and renewable energy technologies (mainly hydropower) already make up a large share of total power supply in Africa (Ouedraogo, 2017, van der Zwaan et al., 2018). Many countries are actively developing their renewable energy resource potential. At the same time, almost 30% of global oil and gas discoveries made in recent years have been in sub-Saharan Africa, reflecting the growing global demand for African fossil fuel resources (IEA, 2014).

To better understand these developments and the trade-offs involved, we develop in this article a comprehensive scenario analysis of the energy and emissions transitions inherent to the rise of the energy sector in Mozambique. The IEA expects that Mozambique will soon join the group of leading energy producers in Africa, together with Nigeria, South Africa and Angola (IEA, 2014). Like much of Africa, Mozambique's natural resources include both renewable and non-renewable energy reserves. In its energy revolution, development of the country's

enormous fossil fuel reserves competes with development of its unusually large hydropower potential. Hence, the case of Mozambique exemplifies the future development of Africa's energy sector in many respects. Insight into the case of Mozambique, therefore, transcends local and regional importance.

Our scenario analysis starts at 2015 and subsequently covers the country's anticipated surge in natural resource exploration until 2030. We develop four basic scenarios to explore the expected impact of this natural resource boom on energy supply and demand as well as on emissions from energy production, transformation and energy demand, differentiated by source, sector and destination (export vs national demand). In addition, using the structure of these four basic scenarios, we develop various sustainable energy policy scenarios that are in line with climate change mitigation and adaptation strategies as recently presented by the Government of Mozambique. Until very recently, Mozambique's contribution to global greenhouse gas (GHG) emissions has been minimal, due to its comparatively small population size in combination with the relatively small size of its underdeveloped economy. This situation, however, is changing rapidly because of the country's natural resource boom, in conjunction with high population growth and a rapidly expanding economy. Through its Intended Nationally Determined Contribution (INDC) to the UN Framework

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Convention on Climate Change, Mozambique aims to reduce emissions with 67,2 MtCO₂eq in the period 2020–2030. Against this background, our energy policy scenarios assess how the emission pathways in the basic scenarios change if the country implements the key policy measures to promote low-carbon development as proposed in its National Climate Change Adaptation and Mitigation Strategy (ENAMMC) (Government of Mozambique, 2012).

Our scenario analysis makes use of MOZLEAP, the first integrated long-run scenario model of the Mozambican energy sector, which we recently developed (Mahumane and Mulder, 2016). The model is based on the Long-range Energy Alternatives Planning System (LEAP), an integrated modeling tool that can be used to track energy consumption, production and resource extraction in all sectors of an economy (Heaps, 2012). In Mahumane and Mulder (2016), we introduced and described the model in detail but only included a minimal energy scenario analysis that merely served the aim of illustrating the calibration and potential of our model; we focused on a few key aggregate trends, we did not pay attention to emissions and did not assess energy policies. In contrast, this article uses the MOZLEAP model to develop a comprehensive review of the energy and emission future of Mozambique, as well as the impact of a range of sustainable energy policies on the country's energy and emission pathways.

Our analysis fits in the literature of LEAP-based studies that present energy and emission scenarios at the country level. Recent examples include studies on China (Wang et al., 2011; Zhou et al., 2013), Greece (Roinioti et al., 2012), Japan (Takase and Suzuki, 2011), Nigeria (Emodi et al., 2017) and Taiwan (Huang et al., 2011; Yophy et al., 2011). Also, LEAP has been used for sector-level analysis in a country or region, often focusing on the power sector (Bautista, 2012; Kale and Pohekar, 2014; McPherson and Karney, 2014) but also on renewable energy planning (Jun et al., 2010). Ouedraogo (2017) uses LEAP to develop long-term scenarios of Africa's energy future, with an emphasis on projecting energy demand and related emissions. In addition, Dalla Longa and Van der Zwaan (2017) use the TIMES-based TIAM-ECN integrated assessment model to analyze the role of renewable and other low-carbon energy technologies in the power sector and on the demand side to achieve Kenya's climate change mitigation targets. Lucas et al. (2015) discuss long-term energy-system developments in Africa, based on a model inter-comparison study on global climate policy. Amongst others, they find a considerable spread in model outcomes, emphasizing the large uncertainty regarding Africa's energy future but also that the period before 2050 is critical for the transition towards a low carbon future. Van der Zwaan et al. (2018) recently examined the prospects for the large-scale use of low-emission energy technologies in Africa from an energy-system cost minimization perspective. In contrast to these continent-wide analyses, we develop – like Emodi et al. (2017) and Dalla Longa and Van der Zwaan (2017) – a country-specific scenario-based analysis to explore Mozambique future energy and emissions pathways against the background of the country's climate change mitigation ambitions.

The structure of the article is as follows. In Section 2 we briefly describe our model, data and scenario outlines. Sections 3 and 4 present the outlook of energy and emission transitions at the energy sector supply and demand side, respectively. Section 5 presents the results of our sustainable energy policy scenario analyses. Section 6 presents the main conclusions and policy implications of the article

2. Model, data and scenarios

2.1. Model and data

Given the LEAP accounting framework, our model essentially calculates energy and emission trajectories as the product of activity levels (such as GDP, population, physical production levels) and energy or emission intensity per unit of activity. Our scenario analysis has its roots in a (newly collected) rich dataset with information on economic

activity and energy demand per sector, demographic and urbanization trends as well as bottom-up information about mining projects and future electricity distribution. The model is run to test its ability to replicate data for the period 2000–2015 and subsequently to generate multiple forward-looking scenarios for the period 2015–2030. Most of the energy statistics for Mozambique that we use in our analysis were collected and processed by the Directorate of Studies and Planning (DEP) of the then Mozambican Ministry of Energy (ME, 2012).¹

As previously mentioned, the model has been introduced and described in detail in Mahumane and Mulder (2016). For this reason and because of space constraints, we limit ourselves here to a summary of the key mechanisms in our model.² On the supply side, we model electricity production, gas exploration, coal mining and heavy sands mining on a project basis. On the demand side, we model energy demand by households, transport and extractive industries, as well as the agriculture, manufacturing, services, government and other sectors. Also, we specify electricity demand from neighboring countries in the region (especially South Africa), given their essential role in developing the Mozambican electricity market. Energy demand modeling is based on a combination of historical energy and activity level data that we collected as well as information on demographic and urbanization trends supplied by external sources, locally collected bottom-up information about future electricity distribution and GDP elasticities with respect to biomass consumption, sector structure and vehicle ownership. These elasticities were obtained by developing and calibrating three sub-models within LEAP's accounting framework, as follows.

2.1.1. GDP

To model future GDP paths, we start with historical GDP growth rates, excluding the extractive sector, which we label baseline growth. Subsequently, for our scenario period, we assume that baseline growth $g_t^{Y^{Base}}$ follows a logistically declining trend as a function of time t , such that baseline GDP is defined as

$$Y_t^{Base} = Y_{t-1}^{Base} \left((1 + g_{t-1}^{Y^{Base}}) e^{-\delta t} \right), \quad (1)$$

with δ a parameter that determines the speed of decline in the logistic growth curve. The value of δ in Eq. (1) is scenario-specific and chosen such that annual GDP growth from 2016 gradually evolves towards 3.8–5.6% by 2030, depending on the basic scenario (see Table A.1 in the Supplementary Material). Next, using a bottom-up approach, we construct GDP separately for each extractive industry project. Total Extractive GDP is then calculated as the aggregate result of expected price trends³ $g_{i,t}^{p^{Extr}}$ and physical production patterns per project i , such that

¹ Underlying data have been provided by a range of local institutions. Economic data on Mozambique were obtained from the National Statistics Institute (INE), the then Ministry of Planning and Development (MPD) and the Mozambique Central Bank (BM) as well as from the World Bank, the International Monetary Fund (IMF, 2013, 2016), the United Nations Department of Economic and Social Affairs (2011 Revision), and the African Development Bank (AfDB). All locally collected data, insofar possible, have been checked against data from international sources, including British Petroleum (BP Statistical Report, 2012), International Energy Agency (IEA, 2013a, 2014), United States Geological Survey (Yager, 2012), United Nations Populations statistics and the World Development Indicators as published by the World Bank.

² Consequently, the text in Section 2 is largely based on Mahumane and Mulder (2016).

³ These prices are partly based on expert judgments for the upcoming years, published in a variety of resources (IEA, 2013b; KPMG, 2013). See Table A.2 in the Supplementary Material for details.

$$Y_t^{Extr} = \sum_i Q_{i,t}^{Extr} [p_{i,t-1}^{Extr} (1 + g_{i,t}^{p^{Extr}})]. \quad (2)$$

Together with baseline GDP this sums up to total GDP:

$$Y_t^{Total} = Y_t^{Base} + Y_t^{Extr}. \quad (3)$$

This implies that total GDP growth rate is a weighted composition of baseline and extractive GDP growth rates. Finally, we assume that sector shares S of the four main sectors (agriculture, services, manufacturing and government) evolve over time as a function of per capita GDP, according to the following logistic curve:

$$S_{(t)} = S_{t-1} * \left[1 + \frac{\theta}{y_t^{Base}} \right]^{\Delta y_t^{Base}} \quad (4)$$

with parameter θ signifying the elasticity of the change in the sectoral composition of the economy under the influence of economic development. The value of θ is sector-specific and is derived from cross-country regressions of the relation between per capita GDP and the respective sector share, using World Bank data for 39 countries with per capita GDP values between US\$700 and US\$3000; estimated coefficients vary from -2.94 for agriculture to 4.86 for manufacturing.

2.1.2. Biomass

We model biomass consumption by adopting a nested model structure (Barnes and Floor, 1999; Barnes et al., 2005). First, we assume that total biomass consumption is merely determined by GDP per capita, thus considering substitution with modern energy forms (such as LPG and electricity) as a function of relative prices to be a second-order effect (Leach, 1992). Second, we assume that the choice of one of the two dominant forms of biomass (fuelwood and charcoal) is driven by their relative prices as well as the urbanization rate (see Eq. (6)). More specifically, we define the evolution of per capita biomass consumption B over time t according to a logarithmic S-shaped curve, as follows:

$$B_t = \alpha [1 + \beta e^{-\gamma y_t^{Total}}], \quad (5)$$

where α is the initial value of B (in the year 2000), β is a constant (vertical shift of the curve), and γ the elasticity of B with respect to GDP per capita y . The value for α is estimated based on a combination of international data (IEA, 2010) and local household survey data (Atanassov et al., 2012; INE, 2009) and equals 10.5 GJ per capita. The values for β and γ are derived from a cross-country logarithmic panel data regression of biomass consumption on per capita GDP for the period 1971–2006, using IEA data for 74 countries with per capita values below US\$3000; estimated coefficients for β and γ equal 0.0274 and 0.239 , respectively. Next, we define the evolution of per capita consumption of charcoal C and fuelwood F as follows:

$$C_t = B_t \lambda_t \text{ with } \lambda_t = \lambda_{t-1} [(1 + \rho)\gamma] \quad (6)$$

$$F_t = B_t [1 - \lambda_t] \quad (7)$$

where λ is the share of charcoal in total biomass consumption, ρ is the inter-fuel substitution elasticity (i.e. between charcoal and fuelwood) and γ is the annual growth rate urbanization. Historical values for γ were derived from census data (INE, 2010), whereas values for λ in the initial year (2000) and ρ were derived from local household survey data (Atanassov et al., 2012; INE, 2009), with ρ set at 0.03 . Future values for γ are taken from expected urbanization trends published by the UN in its World Urbanization Prospects (UN, 2008). Building on these assumptions we model future evolution of biomass consumption per electrified household b_t^{Elec} as a function of changes in the total biomass consumption (Eqs. 5–7) as well as the change in urbanization rate U relative to the change in the electrification rate E , according to

$$b_t^{Elec} = \Delta b_{t-1} \left[1 + \frac{\Delta U}{\Delta E} \right], \quad (8)$$

with b representing either charcoal C or fuelwood F per electrified

household. Biomass intensity per non-electrified household is subsequently derived from total biomass consumption not consumed by electrified households.

2.1.3. Transport

We model fuel demand for road transport as a function of vehicle ownership over time, given the evolution of per capita GDP and population. We assume the number of vehicles per 1000 people to be merely determined by baseline GDP per capita, thus considering (relative) fuel prices a second-order effect. More specifically, we define the number of vehicles V per 1000 people at time t according to:

$$V_t = V_{t-1} * \left[1 + \frac{\psi}{y_t^{Base}} \right]^{\Delta y_t^{Base}} \quad (9)$$

with parameter ψ denoting the elasticity of the change in vehicle ownership under the influence of economic development. The value of ψ is derived from a cross-country logarithmic panel data regression of passenger car ownership on per capita GDP for the period 1971–2006, using data from the World Bank Indicators database for 86 countries with per capita values below US\$3000; the estimated coefficient for ψ equals 8.7 . We calibrate this function based on data on the annual evolution of registered vehicles in Mozambique and fuel consumption in the recent past, supplied by, respectively, the Mozambican National Institute of Road Transport (INATTER, 2012) and the then Ministry of Energy (ME, 2012).

2.1.4. Exports

In Mahumane and Mulder (2016), we have shown that, due to the combination of sizeable future production and a relatively small domestic economy, we expect that 80–90% of total energy production in Mozambique is expected to be destined for export. This includes natural gas and coal production for the world market, but also electricity production for the regional market of Southern Africa. Future electricity capacity is anticipated to meet the combined effect of increased power demand from the Southern African power pool (SAPP), a range of coal and heavy sands mining projects, and other domestic demand. In LEAP we model South Africa separate, because of its key role in Mozambique's electricity sector, while exports to the rest of SAPP are modeled as one category. Projections of regional demand for Mozambique's electricity production are based on information from IRENA (2013), Eskom's Integrated Resource Plan for Electricity (SA Department of Energy, 2011), a recent study by Spalding-Fecher et al. (2017) and interviews with local experts.

2.2. Scenarios

As described in Section 1, we first develop four basic scenarios to present a consistent picture of how the country's energy system may evolve over time under the influence of different natural resource exploration trajectories, and realistic expectations of GDP and population growth. The first three of these basic scenarios—which we label ‘Reference’, ‘Reference High’ and ‘Reference Low’—are largely defined by GDP and population growth expectations. Reference is the most likely development path, in which GDP growth is based on baseline projections plus activities of new extractive industry and electricity generation projects that are (almost) sure to be realized, considering realistic and somewhat conservative estimates about the output price development in the extractive and aluminum industry. Furthermore, it adopts a medium variant of population growth scenarios, a modest decline in household size, a moderate speed of urbanization and somewhat conservative estimates about the development of energy intensity improvements across sectors. Reference High and Reference Low then refer, respectively, to the optimistic and pessimistic variant of Reference, thus assuming higher (lower) baseline economic growth, lower (higher) population growth, higher (lower) speed of urbanization,

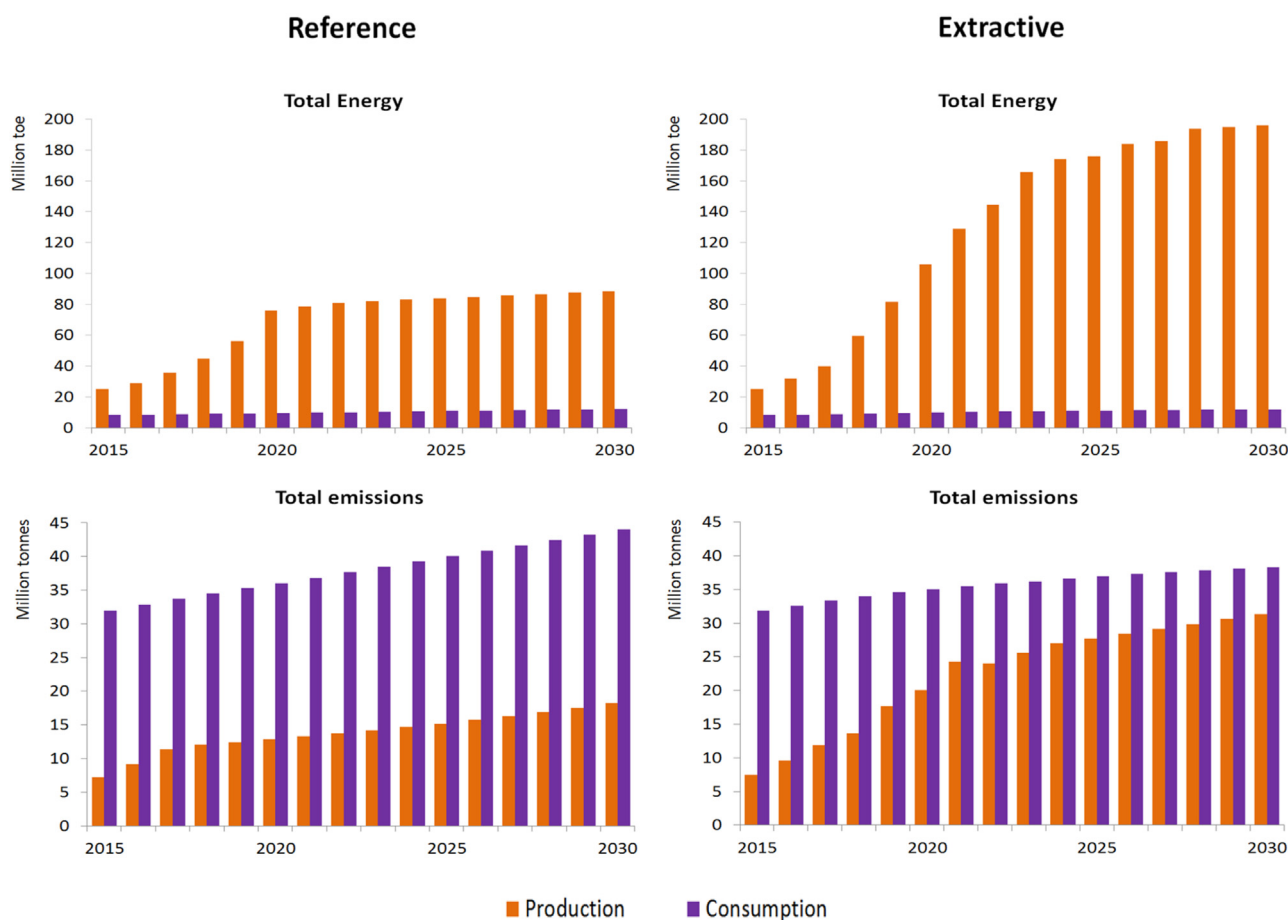


Fig. 1. Primary energy production vs final energy consumption (top), and total emissions from energy production and consumption (bottom) – Reference and Extractive scenario.

faster (slower) decline of household size and higher (lower) output price developments in the extractive and aluminum industry.

The fourth basic scenario—which we label ‘Extractive’—describes the expected evolution of the Mozambican energy system if all potential projects in power generation and the extractive and aluminum industries are realized, including those projects that are yet (very) uncertain. In other words, this scenario tells the story of the Mozambican economy and energy sector becoming very much extractive industry-driven. Because of this focus, we assume all other leading dimensions of the model (population growth, household size, speed of urbanization, energy intensity improvements and output price developments) to be equal to the Reference or Reference High scenario. We refer to [Table A.1 in the Appendix](#) for more details on the key scenario assumptions.

To illustrate the role of the basic scenarios in defining the energy and emission pathways of Mozambique, we present in [Fig. 1](#) the evolution of total energy and emissions levels over time for the Reference scenario (left) and the Extractive scenario (right) in terms of both energy production and energy consumption.⁴ These graphs thus outline, respectively, the most likely development path and the path that assumes exploitation of Mozambique’s natural resource reserves to its fullest potential.

The top of [Fig. 1](#) shows that in the period 2015–2030 primary energy production is likely to increase around 4 to 8-fold, from 25 million tons of oil equivalence (toe) in 2015–88 million toe in 2030 in the Reference case. In the Extractive scenario, this could even increase to over 196 million toe in 2030. Indeed, this means that Mozambique will

undergo a revolution at the supply side of its energy sector. In contrast, energy consumption is expected to increase by about 50% over the same period, from about 8 million toe to about 12 million toe. The bottom of [Fig. 1](#) shows that the opposite pattern holds for total emissions, with emissions from energy consumption being much larger than emissions from energy production. At the same time, it is expected that emissions from energy production will grow much faster over time. In short, total emissions from energy consumption are expected to increase with 23–42% over a 15-year period: from 31 million tonnes in 2015–44 million tonnes in 2030 in the Reference scenario, and 38 million tonnes in the Extractive scenario. In contrast, we foresee that in the same period total emissions from energy production increase by about 160–340%: from 7 million tonnes in 2015–18 or 31 million tonnes in, respectively, the Reference and Extractive scenario.

The relative stagnation of energy production growth in the Reference scenario is mainly caused by stabilized production quantities of natural gas, after a few years of rapid expansion of natural gas production capacity. In addition, in the Extractive scenario the aluminum smelter MOZAL – in 2015 still responsible for about 60% of total electricity demand in Mozambique – is expected to expand its production capacity in 2020, implying a 30% increase of its electricity demand (see [Section 4](#)). Hence, growth rates of GDP, energy supply and electricity demand in Mozambique are expected to peak around 2020, and to be much lower afterwards. Consequently, the ‘stagnation’ that can be found in several figures in this study should be merely interpreted as a rapid transition to a higher steady state in terms of GDP level, energy production and electricity demand – including the corresponding emissions (see [Figs. 3–5](#)).

In [Sections 3 and 4](#) we decompose these aggregate trends into their

⁴ Note that in terms of production, the Reference Low and Reference High scenarios are (very) similar to the shown Reference scenario.

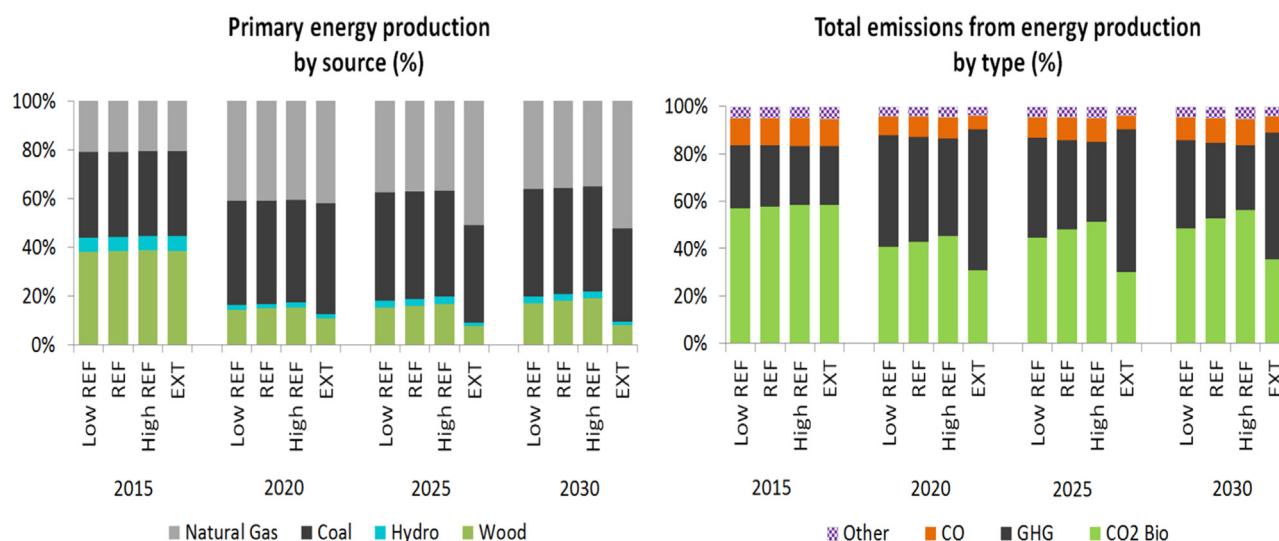


Fig. 2. Decomposition of total primary energy production by source (left) and total emissions from energy production by emission type (right).

underlying determinants to gain insight into how the energy and emissions transitions in Mozambique may evolve until 2030, without additional and substantial climate change mitigation and adaptation measures. Next, we will explore in Section 5 how these energy and emission pathways scenarios change if the country would implement low-carbon development policies as proposed in its National Climate Change Adaptation and Mitigation Strategy (ENAMMC) (Government of Mozambique, 2012). To this end, we develop seven energy policy scenarios based on three supply-side policy options in the electricity subsector and four demand-side policy options. All these energy policy scenarios build upon the structure of the four basic scenarios described above.

3. Supply-side transitions

As noted in the previous sections, until 2030 Mozambique is expected to undergo an energy revolution in terms of (primary) energy supply, driven by exploration and transformation of its abundant natural resource stock. In Fig. 2 we decompose this aggregate trend into energy production levels per energy type (left) and emissions from energy production per energy type (right), across all our four basic scenarios.⁵ At the turn of the century, 80–90% of total energy production in Mozambique consisted of wood, with hydro largely making up for the remaining 10–20%. The left-hand side of Fig. 2 shows, however, that from 2020 onwards natural gas and coal together will make up 80–90% of aggregate production levels. Nevertheless, our scenarios indicate that total energy production from wood and hydro are expected to increase 60–90% over the next 15 years; yet the emergence of natural gas and coal are causing their relative shares in total energy production to decrease to about 1–3% (hydro) and 8–19% (wood) in 2030, depending on the scenario.

As regards emissions from energy production, the right-hand side of Fig. 2 shows that over time emissions from energy production are mainly driven by GHG emissions under the influence of fossil fuel-based electricity generation, plus biogenic carbon dioxide emissions resulting

from charcoal making. The latter type of emissions rises most (least) in the Reference High (Low) scenario because in our model we assume that with rising per capita GDP relative charcoal consumption increases under the influence of rising income and urbanization. It is noteworthy that, by 2030 these carbon-neutral emissions are still responsible for roughly 40–50% of total emissions, notwithstanding the unprecedented increase in modern energy production. Fig. 2 also shows that GHG emissions rise most intensely in the Extractive scenario, evidently because this scenario assumes that all potential projects in power generation and the extractive and aluminum industries are to be realized.

To gain further insight in the underlying determinants of the emission transition paths, we present in Fig. 3 for the Reference and Extractive scenarios a decomposition of total GHG emissions from energy transformation, respectively per output fuel (top) and feedstock fuel (bottom).⁶ The top of the Figure shows that emissions from charcoal production and electricity generation tend to become almost equally important in the Reference scenario, while electricity generation is by far the dominant driver of GHG emissions from energy transformation in the Extractive scenario. The bottom of Fig. 3 shows that coal-fired electricity generation is by far the most important source of GHG emissions in the transformation sector, especially in the Extractive scenario. In the Reference and Extractive scenarios, by 2030 coal is expected to be responsible for, respectively, 34% and 61% of total GHG emissions associated with complete energy transformation. If we only look at electricity generation, by 2030 coal accounts for 69% and 78% of GHG emissions associated with electricity generation in the Reference and Extractive scenarios, respectively. Transitions in the electricity sector thus play a crucial role in shaping the country's emerging energy sector and its future levels of global (GHG) emissions.

As noted before, the majority (80–90%) of total energy production in Mozambique is expected to be destined for export—including natural gas and coal production for the world market and electricity production for the regional market of Southern Africa. Electricity production is expected to double until 2030 in the Reference scenario (from about 20 GWh in 2015–37 GWh in 2030), while it could also triple to about 54 GWh if Mozambique were to follow the extractive scenario path. Despite the emergence of fossil fuel-based electricity generation, hydro remains the main source (60–80%) of electricity production in

⁵ We exclude Solar and Diesel from Fig. 4 because their values are too small to visualize. In the right-hand side of Fig. 4 the acronyms are defined as follows. GHG = Greenhouse Gas emissions, sum of Carbon Dioxide (CO₂), Methane (CH₄) and Nitrous Oxide (N₂O); CO₂ Bio = Biogenic Carbon Dioxide; Other emissions include a range of local and regional air pollutants, such as Carbon Monoxide (CO), Non-Methane Volatile Organic Compounds (NMVOCs), Nitrogen Oxides (NO_x), Particulates PM₁₀, Tot Suspended Particulates (TSP), Sulfur Dioxide (SO₂) and Sulfur Oxides (SO_x).

⁶ We focus here on the Reference and Extractive scenarios because of space constraints, while this is justified by the fact that, because of our scenario design, the results for the various Reference scenarios are (very) similar in terms of (emissions from) energy production.

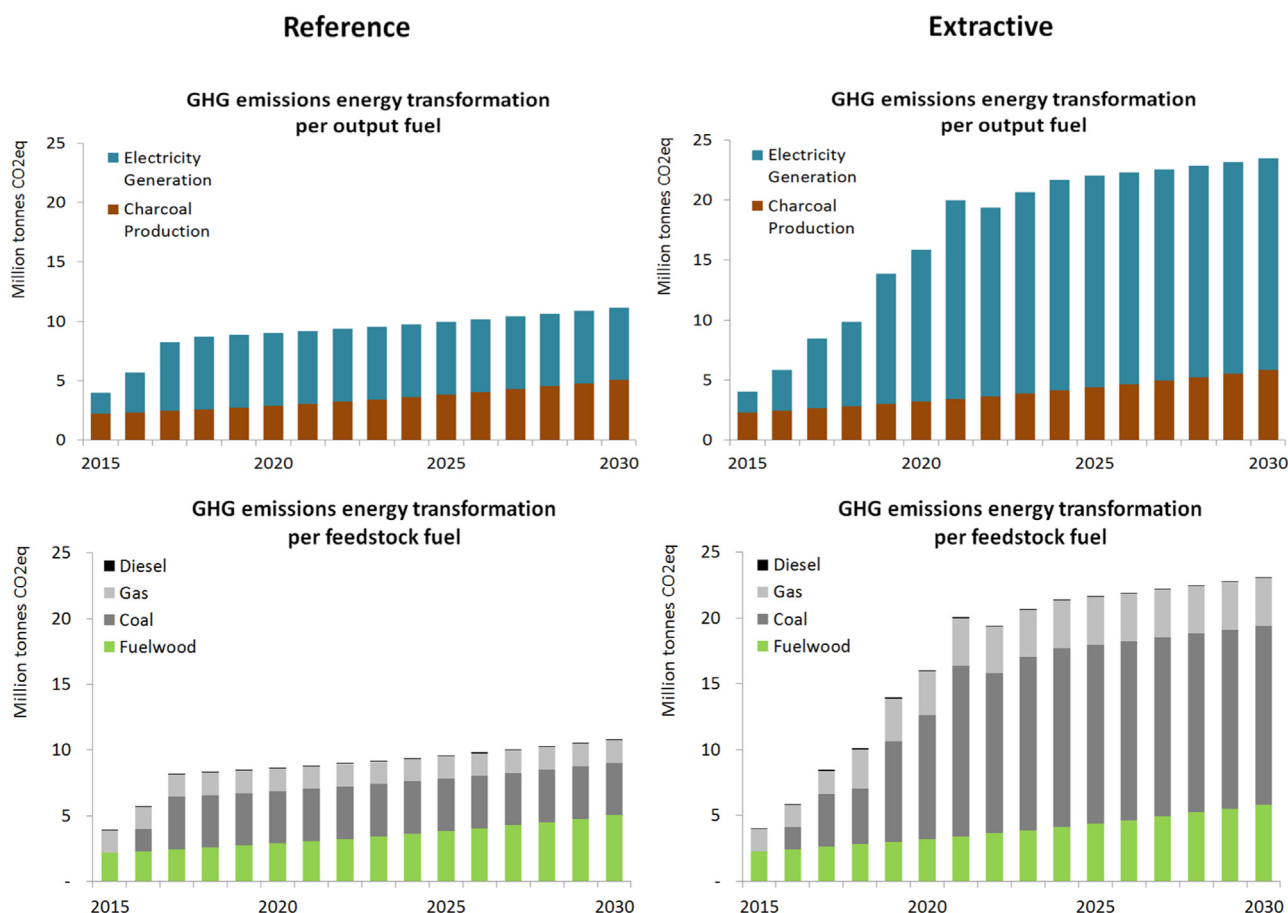


Fig. 3. GHG emissions from energy transformation per output fuel (top) and feedstock fuel (bottom) – Reference and Extractive scenario.

Mozambique—because of the existing Cahora Bassa Hydroelectric (HCB) dam in combination with the planned future construction of other major hydro projects. About half of the expected total electricity generation capacity is needed to meet domestic electricity demand, while the other half thus serves electricity demand from neighboring countries, especially South Africa.

In Fig. 4 we show the emission implications of this production structure. In the Reference and Extractive scenarios, energy production for export is responsible for, respectively, 30% and 35% of the GHG emissions associated with energy production. The remaining emissions share can be linked to energy production for the domestic market, with

a small margin for which the destination of future energy (electricity) production is yet unknown.

4. Demand-side transitions

Households are responsible for a major part of total energy consumption in Mozambique (about 90% in 2000 and 50–60% by 2030), followed by the transport sector and industry (Mahumane and Mulder, 2016). Our scenario model assumes that a rising GDP is inversely related to population growth (scale effect) while raising incomes and thereby increasing energy demand (income effect) but also inducing a

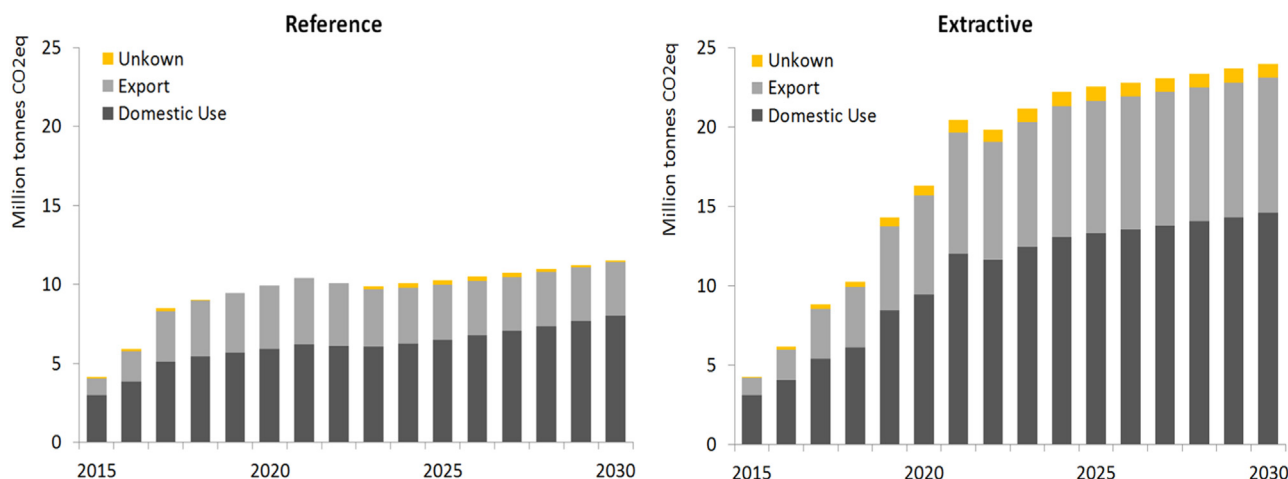


Fig. 4. GHG emissions from energy production by destination – Reference and Extractive scenario.

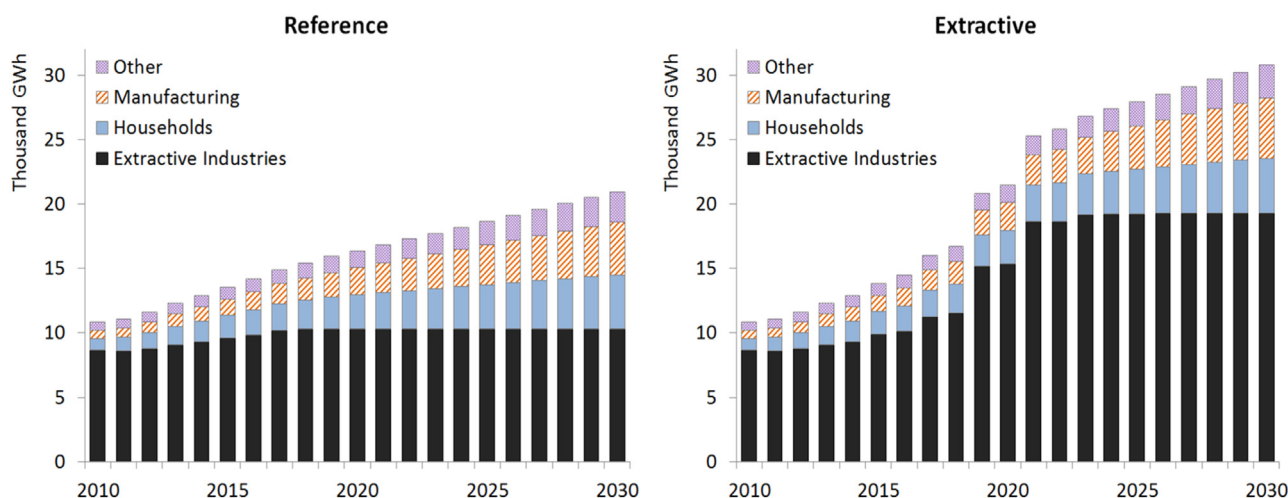


Fig. 5. National demand for electricity by sectors – Reference and Extractive scenario.

substitution away from consumption of biomass towards modern energy forms (efficiency effect). Our calculations show that the scale effect is the main driver of future demand, while the intensity effect outweighs the income effect. Consequently, long-run aggregate energy demand is considerably higher in the Reference Low scenario than in the Reference High and Extractive scenario. Across all scenarios, the role of urbanization, which impacts LPG use and electrification, is relatively small. Energy demand by the non-residential sectors obviously increases most in the Extractive scenario because of the assumed growth of the extractive and manufacturing sectors. Finally, demand by road transport varies only to a relatively limited extent among the various scenarios and features an average annual growth of more than 7%; this implies that demand for fuel transport is expected to increase by more than 2.5 times between 2015 and 2030.

Economic growth, continued electrification and extractive industry expansion lead to a substantial increase in electricity consumption until 2030. In Fig. 5 we present a decomposition of the evolution of Mozambique's domestic electricity consumption over time—again showing only the Reference and Extractive scenarios because of their distinct patterns.

The breakdown of domestic electricity demand in Fig. 5 leads to various observations. First, the extractive sector continues to dominate the picture of national electricity consumption. This result is mainly driven by the aluminum smelter MOZAL, which transforms imported bauxite into aluminum for export, facilitated by cheap electricity. In the

Extractive scenario MOZAL is expected to expand its production capacity around 2020, implying a 30% increase of its electricity demand. In addition, we expect that by 2030 the combined electricity consumption from other extractive industry firms has grown to a level of 2.3 thousand GWh in the Reference scenario and 8.5 thousand GWh in the Extractive scenario, most notably because of heavy sands mining. Second, Fig. 5 shows a rapid growth in electricity demand for manufacturing and household use, while the demand trend by the other sectors (Services, Government, Agriculture and other) is more gradual. Total electricity consumption by households more than doubles under the influence of increasing (urban) welfare and (rural) electrification, to reach 4.2 thousand GWh by 2030 in both scenarios. In comparison, by 2030 manufacturing electricity consumption is expected to be 3–4 times higher than in 2015.

Fig. 6 shows the energy mix and emission implications of these trends. The left-hand side of the Figure first shows that traditional biomass consumption in Mozambique continues to play a dominant role in the energy mix—in absolute terms as well as with respect to its role in explaining differences across the scenarios. More precisely, in the Reference scenario, traditional biomass consumption (fuelwood plus charcoal) remains responsible for about 60% of total final energy consumption by 2030. Even in the high economic growth scenarios (Extractive and Reference High), about 50% of total final energy consumption in 2030 will consist of traditional biomass, while in the low growth scenario (Reference Low) this percentage is 63%. Second, Fig. 6

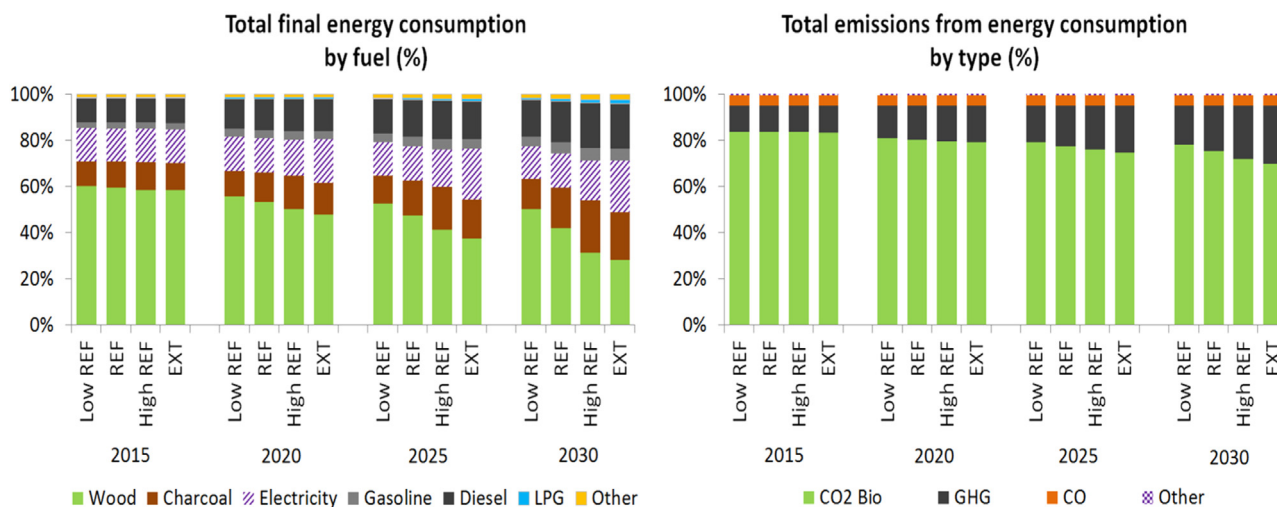


Fig. 6. Total final energy consumption by fuel (left) and emissions from energy consumption by type (right).

shows that the spectacular growth figures for electricity consumption (see Fig. 5) do not translate to a strong increase of electricity's share in total final energy consumption because of the emergence of other energy types, most notably Diesel and Charcoal.

The right-hand side of Fig. 6 provides a decomposition of total emissions from energy consumption by different emission types.⁷ It clearly shows that biogenic CO₂ emissions, which mainly result from traditional biomass consumption by households, is by far the dominant driver of the observed aggregate level and growth rate of total emissions. In short, depending on the scenario, by 2030 Biogenic CO₂ emissions account for 70–75% of total consumption-driven emissions, against 83% in 2015. In comparison, by 2030 GHG emissions account for 20–25% of total consumption-driven emissions, against 12% in 2015.

In Fig. 7 we provide a sectoral breakdown of consumption-driven GHG emissions. It shows that by 2030 GHG emissions from increased demand for (urban) transport by car account for two-thirds of total consumption-driven GHG emissions, up from 55% in 2015. The second-largest source of consumption-driven GHG emissions by 2030 is the household sector: 17–20%, down from almost 34% in 2010. Finally, the decomposition of non-transport related GHG emissions by energy source presented in the lower part of Fig. 7 shows that fuelwood consumption continues to be also an important driver of GHG emissions—in the form of methane and nitrous oxide—although this importance is expected to decrease over time with rising GDP levels. In contrast, over time GHG emissions from energy consumption are expected to be increasingly driven by consumption of LPG, charcoal, and especially natural gas and diesel by households and manufacturing. In sum, these trends illustrate the gradual transition away from consumption of traditional biomass and towards consumption of fossil fuels.

5. Emission effects of energy policies

We conclude this study by analyzing how the emission pathways in the basic scenarios, as presented in the previous sections, change if Mozambique implements energy policies to promote low-carbon development. The Government of Mozambique (GoM) has officially committed itself to reducing emissions by 76,5 MtCO₂eq in the period 2020–2030, as stated in its Intended Nationally Determined Contribution (INDC) to the UN Framework Convention on Climate Change in Paris (Government of Mozambique, 2015). This goal was the first ever quantitative national emission target set by the GoM. Because Mozambique's contribution to global greenhouse gas (GHG) emissions has, until recently, been very limited, low-carbon strategies are a relatively new policy area in Mozambique. The INDC emission reduction goal is an estimate based on policy actions and programs outlined in a series of national strategies that have been developed by the GoM over the last years.⁸ Recently, the goal was updated to 67,5 MtCO₂eq,⁹ while still being conditional on “the provision of financial, technological and capacity building from the international community” (Government of Mozambique, 2015:9). Furthermore, it is noteworthy that, in the series

of energy-related strategy documents by the GoM that support this goal, the vast majority of policies proposed are described in very general terms, without quantitative targets.¹⁰

Given these uncertainties, in developing our energy policy scenarios, we decided to first choose four of the most prominent energy demand policies proposed in the various national strategy documents and to assess these policy options in terms of their marginal impact on emissions by assuming a 1% change from the respective basic scenarios for the period 2018–2030. These four policy options are: i) a 1% increase in LPG consumption by households at the expense of a 1% decrease in charcoal consumption; ii) a 1% increase in the share of passenger cars that runs on natural gas; and 3) a 1% increase in the efficiency of traditional biomass consumption, for example by the distribution of improved stoves; and iv) a 1% efficiency increase of energy consumption by the main non-extractive sectors in the economy—manufacturing, services and government. Next, we look at the supply-side of the energy system, with the aim of assessing the expected emission impact of developing Mozambique's large renewable electricity potential against the planned expansion of thermal electricity generation capacity based on coal and natural gas. Although construction of specific power plants is part of the energy strategy of the GoM, the trade-off between large-scale exploration of renewable and fossil fuel reserves for electricity generation is not explicitly made. Therefore, we have opted to make this trade-off transparent by evaluating the impact of three radical sustainable energy policy options in the electricity subsector that could change the course of Mozambique's energy sector: i) a turn towards 100% renewable energy for electricity based on hydropower only; ii) a ban on coal-fired power stations; and iii) a substantial increase of solar energy in the electricity mix at the expense of coal-based electricity generation capacity.¹¹

Clearly, our demand-side policies reflect the political reality, whereas our supply-side policies are merely meant to stimulate the ongoing debate in Mozambique on the necessity and desirability of its dominant economic strategy that is based on energy-related, export-oriented, capital-intensive ‘megaprojects’. Over the last 20 years the ensuing pattern of economic growth in Mozambique has become progressively less inclusive, with the energy-related megaprojects having generated a few jobs directly and indirectly (Lachler and Walker, 2018). As noted in Section 2.2, all our energy policy scenarios build upon the structure of the four basic scenarios that we described in Sections 2–4. The energy policy scenarios are thus modeled together with the basic scenarios to assess how the emission pathways in the basic scenarios change if the country implements policy measures to promote low-carbon development. For the sake of analytical transparency and comparability, we assume in our scenario analysis that all these policy measures take effect per 2020. In Table 1 we present the cumulative emission reduction impact of the various supply and demand-side energy policies.

The most important conclusion from Table 1 is that in Mozambique sustainable energy policies at the supply-side lead to far higher cumulative emission reductions than sustainable energy policies at the demand-side. This is especially true for global (GHG) emissions. Of course, in contrast to the supply-side energy policy options, our demand-side policies have been designed as marginal interventions, assuming a 1% change from our respective baseline scenarios. The results in Table 1, however, show that even a very substantial intervention (let's say

⁷ We refer to footnote 4 for definitions of the acronyms used in the right-hand side of Fig. 6.

⁸ They include the Energy Strategy 2015–2024, the New and Renewable Energy Development Strategy 2011–2025 (EDENR), Strategy for Conservation and Sustainable Use of Biomass, and the National Strategy for Reducing Emissions from Deforestation and Forest Degradation (REDD+). The key policy measures to promote low-carbon development in Mozambique, as proposed in these national strategy documents, are rooted in Mozambique's National Climate Change Adaptation and Mitigation Strategy (ENAMMC) (Government of Mozambique, 2012).

⁹ Internal presentation by Telma Manjate (May, 2018), UNFCCC national focal point, Ministry of Land Environment and Rural Development (MITADER-DINAB).

¹⁰ Exceptions include targets on transmission losses, the electrification rate and construction of power plants and refinery capacity (Energy Strategy 2015–2024); increase in Hydro capacity and number of water pumps and heaters on renewable energy (New and Renewable Energy Development Strategy 2011–2025); emission reduction from lower charcoal production (Strategy for Conservation and Sustainable Use of Biomass).

¹¹ We assume that under policy regime iii) total solar capacity increases to 300 MW by 2030 at the expense of coal-based electricity generation capacity. This equals an average annual increase in solar capacity of almost 20%.

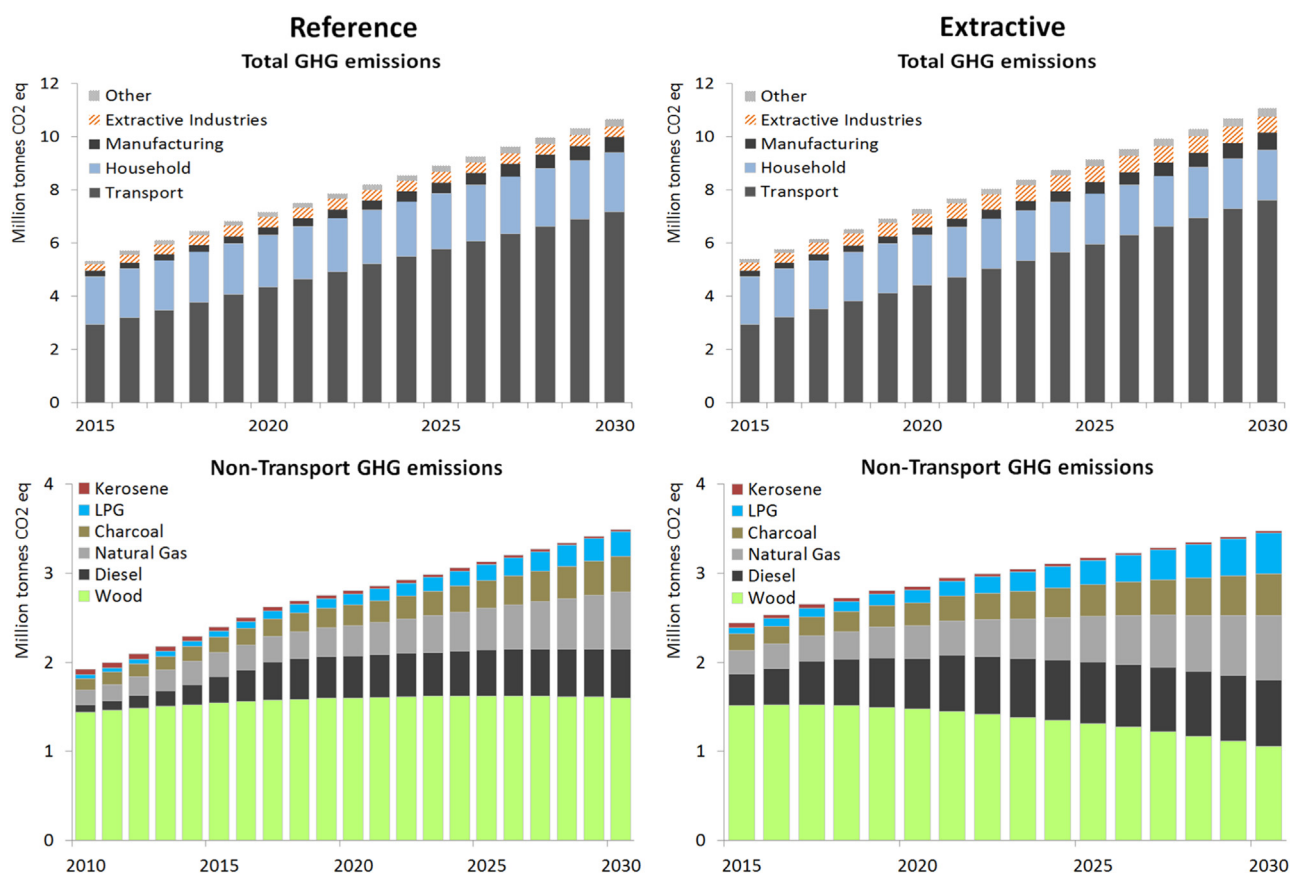


Fig. 7. Total GHG emissions from energy consumption by sector (top) and total GHG emissions for Non-Transport by fuel type (bottom)—Reference and Extractive scenario.

energy efficiency improvements of 25%) will by no means come close to the emission savings potential at the supply-side. In short, the upper part of Table 1 shows that radical choices at the supply-side of the energy system of Mozambique could generate until 2030 a cumulative total GHG emission reduction between 25 million and 180 million ton. The results for the various demand-side policies confirm that in terms of total emissions the highest emission reduction levels can be achieved by reducing consumption of traditional biomass, given the dominant role of traditional biomass in total energy consumption (see Figs. 6 and 7). As such, our analysis supports the notion that reduction of local (indoor) pollution inherent to traditional biomass consumption should have high priority in designing energy policies for developing countries. In terms of GHG emission, the biggest gains clearly can be achieved in the transport sector, given the strong growth in vehicle use (see Fig. 7).

To weigh these emission reductions in the context of Mozambique's Climate Change Adaptation and Mitigation Strategy (ENAMMC), we present in Table 2 the evolution of the emission (reduction) levels in the four basic scenarios as well as the seven energy policy scenarios in terms of Global Warming Potential (GWP), as compared to the emission reduction target set by the GoM in its INDC. The upper part of the table shows that Mozambique's updated emission reduction target of 67.2 MtCO₂eq by 2030 is equivalent to a reduction of almost 30% in comparison to the basic Reference paths, and a 18% reduction in comparison to the Extractive development path. The results for the supply-side energy policies, as presented in the middle part of Table 2, show that this goal is very ambitious unless Mozambique follows the Extractive development path that assumes exploitation of Mozambique's natural resources to its fullest potential. In the latter case, electricity generation using hydro only or a ban on coal-fired power is expected to save about 150–190 MtCO₂eq by 2030—multiple times the INDC target set the GoM. However, in the more realistic Reference scenarios, these radical

interventions would not save enough GHG emissions to reach the INDC target. Furthermore, the results show that a strategy to substantially increase solar energy in the electricity mix would at best lead to a 12% reduction in GWP. Finally, the bottom part of Table 2 shows the marginal reductions in GWP for the various demand-side policies. The last column shows that a 1% efficiency improvement or substitution effect at the demand side leads to a 0.01–0.30% reduction in GWP. These relatively low elasticities are caused by rebound and substitution effects and underline that the most effective low-carbon strategies in Mozambique should target the supply-side of its energy system, in particular the electricity generation subsector.

Since our supply-side policies (partially) phase out thermal electricity generation capacity, they lead to a decrease in electricity supply as compared to the no-policy scenarios. If this leads electricity supply to fall short of future demand, these energy policies would gravely harm future economic development and are thus unlikely to be seriously considered as part of any future energy strategy. Therefore, we calculated for each of the energy policies the reserve margin in electricity generating capacity over time—the difference between supply and demand for electricity—across the various scenarios. The results of this exercise indicate that, if we do not consider electricity exports, the reserve margin will never be lower than 52% and can be as high as 133%. However, if we do consider future foreign demand for electricity generated in Mozambique, the percentage reserve margin reduces at best to a narrow range of 2–14% but will also be negative at certain time periods with production capacity falling short at a range of 5–22%, particularly in the period around 2020 and in case of the 'hydro only' and 'no coal' policy options (see Table A.3 in the Supplementary Material for detailed results). In other words, none of our supply-side energy policies would even come close to threatening future domestic electricity demand and even if Mozambique imposes a complete ban on

Table 1
Cumulative emission savings of various sustainable energy policies, 2020–2030.

Supply-side policies	Million Metric Tonnes					%				
	GHG	Biogenic CO ₂	CO	Other	Total	GHG	Biogenic CO ₂	CO	Other	Total
Hydro only										
Reference [*]	50.7	–	0.1	0.9	51.7	31.4	–	0.2	7.6	7.3
Extractive	176.6	–	0.2	3.5	180.3	62.1	–	0.4	22.4	22.5
No coal										
Reference [*]	44.5	–	0.1	0.9	45.5	27.5	–	0.2	7.4	6.5
Extractive	137.3	–	0.2	3.4	140.8	48.2	–	0.4	21.7	17.5
Solar intensive										
Reference [*]	24.7	–	0.0	0.5	25.2	15.3	–	0.1	4.2	3.6
Extractive	28.0	–	0.2	0.6	28.7	9.8	–	0.4	3.8	3.6
Demand-side policies										
	Thousand Metric Tonnes					%				
	GHG	Biogenic CO ₂	CO	Other	Total	GHG	Biogenic CO ₂	CO	Other	Total
LPG households, 1%										
Reference ^{**}	– 1.9	– 1598.5	– 207.4	– 67.6	– 1875.3	– 0.001	– 0.327	– 0.508	– 0.550	– 0.266
Extractive	5.5	– 1663.9	– 215.9	– 70.3	– 1944.6	0.002	– 0.361	– 0.519	– 0.452	– 0.242
LPG Transport, 1%										
Reference ^{**}	– 63.3	–	– 19.1	– 4.1	– 86.5	– 0.039	–	– 0.047	– 0.033	– 0.012
Extractive	– 65.6	–	– 19.7	– 4.2	– 89.6	– 0.023	–	– 0.047	– 0.027	– 0.011
Biomass efficiency, 1%										
Reference ^{**}	– 32.1	– 4857.0	– 375.4	– 96.6	– 5361.1	– 0.020	– 0.991	– 0.922	– 0.786	– 0.761
Extractive	– 33.8	– 4571.9	– 382.9	– 103.6	– 5092.2	– 0.012	– 0.991	– 0.921	– 0.665	– 0.634
Energy efficiency, 1%										
Reference ^{**}	– 35.0	–	0.0	– 0.1	– 35.1	– 0.022	–	0.000	– 0.001	– 0.005
Extractive	– 39.7	–	0.0	– 0.1	– 39.9	– 0.014	–	0.000	– 0.001	– 0.005

GHG = Greenhouse Gas emissions, sum of Carbon Dioxide (CO₂), Methane (CH₄) and Nitrous Oxide (N₂O); Biogenic CO₂ = Biogenic Carbon Dioxide; Other emissions include a range of local and regional air pollutants, such as Carbon Monoxide (CO), Non-Methane Volatile Organic Compounds (NMVOCs), Nitrogen Oxides (NO_x), Particulates PM₁₀, Total Suspended Particulates (TSP), Sulfur Dioxide (SO₂) and Sulfur Oxides (SO_x).

* Results are identical for all three Reference scenarios.

** Results are average of results across the three Reference scenarios.

thermal electricity generation, it would still be able to export substantial amounts of electricity because of its large hydropower potential.

6. Conclusions and policy implications

This article presented the first comprehensive energy and emission scenario analysis for Mozambique until 2030. Mozambique is rich in natural resources and is rapidly emerging as a leading energy producer in Africa. Its energy revolution features a competition between development of vast reserves of renewable and fossil energy sources. This makes the case of Mozambique to exemplify the energy sector in Africa at large. Our analysis included an evaluation of various sustainable energy policy options in terms of their impact on reducing total and GHG emissions.

Our analysis revealed an emerging ‘energy-dichotomy’ in Mozambique. In the period 2010–2030, primary energy production is likely to increase 4 to 8-fold whereas energy demand increases by ‘only’ 50% over the same period. By implication, most of the country’s future energy production is expected to be destined for export. The expected surge in energy production comes with a drastic change in the energy mix. In 2000, 80–90% of total energy production in Mozambique consisted of wood, while by 2030 80–90% of total energy production consists of natural gas and coal. As a result, the share of modern renewable energy sources in primary energy production becomes marginal, despite a substantial absolute increase in production volumes for wood and hydro over the next 15 years. By 2030 the residential sector still accounts for the major part of total energy consumption in Mozambique, with large parts of the population still depending on traditional biomass to meet their energy needs. However, increasing per capita income and urbanization are expected to incur a shift away from fuelwood to charcoal. In contrast to the energy supply side, we find that

the share of modern renewable energy sources in the energy consumption mix is expected to remain stable over time at around 15%, caused by increasing consumption of hydro-electricity under the influence of a gradually increasing electrification rate.

In Mozambique total emissions from energy consumption outweigh emissions from energy production. The former mainly consist of biogenic carbon dioxide emissions from traditional biomass consumption by households, which is also by far the dominant driver of the aggregate level and growth rate of total emissions in Mozambique. Emissions from energy production mainly consist of GHG emissions from fossil fuel-based electricity generation supplemented with emissions from (coal) mining activities. About one-third of these emissions is driven by Mozambique’s energy export. This of course raises the question as to the role of other countries in determining the development path of the energy system in Mozambique. Clearly, our analysis suggests that without substantial global reductions in the use of carbon intensive forms of energy, Mozambique is in danger to develop as a pollution haven in the global and regional energy market. The continued dominant role of traditional biomass consumption in the energy mix of Mozambique until 2030 (and most likely much beyond) underlines the importance of strengthening the institutional framework – amongst others, supported by the REDD+ (Reducing Emissions from Deforestation and Forest Degradation in Developing Countries) process – to oppose future forest degradation and to secure sustainable woodland ecosystem services in Mozambique (Siteo et al., 2012; Zorrilla-Miras et al., 2018). We think that further reinforcing the link between scenario models for future energy use and land-use trends would be highly beneficial to this process, and a promising area for further research.

Until recently, Mozambique’s contribution to global greenhouse gas (GHG) emissions has been minimal. As a result, policies for low-carbon development are a relatively new policy area in Mozambique. The

Table 2
Global Warming Potential impact of various sustainable energy policies, 2020–2030.

Basic Scenarios	Annual					Cumulative		
	Level					Level	Reduction – INDC goal	
	MtCO ₂ eq					MtCO ₂ eq	MtCO ₂ eq	%
	2015	2018	2020	2025	2030	2018–30	2018–30	2018–30
Reference	9.3	15.2	16.2	18.9	21.8	239.1	– 67.2	– 28.1
Reference High	9.4	15.4	16.6	19.8	23.5	250.1	– 67.2	– 26.9
Reference Low	9.2	14.9	15.8	18.1	20.4	229.1	– 67.2	– 29.3
Extractive	9.4	16.4	23.2	31.1	34.5	371.8	– 67.2	– 18.1
Supply-side policies	Savings compared to basic scenarios					Level	Reduction – scenarios	
	MtCO ₂ eq					MtCO ₂ eq	MtCO ₂ eq	%
	2015	2018	2020	2025	2030	2018–30	2018–30	2018–30
Hydro only								
Reference*	– 0.1	– 4.5	– 4.5	– 4.5	– 4.5	181.3	– 58.2	– 24.3
Extractive	– 0.2	– 5.4	– 11.0	– 17.6	– 16.0	181.3	– 190.6	– 51.3
No coal								
Reference*	0.0	– 4.0	– 4.0	– 4.0	– 4.0	187.5	– 51.9	– 21.7
Extractive	0.0	– 3.5	– 8.6	– 13.6	– 13.6	220.7	– 151.1	– 40.6
Solar intensive								
Reference*	0.0	– 0.4	– 1.0	– 2.5	– 4.0	210.8	– 28.7	– 12.0
Extractive	0.0	– 0.5	– 1.6	– 2.5	– 4.0	339.6	– 32.2	– 8.7
Demand-side policies	Savings compared to basic scenarios					Level	Reduction – scenarios	
	Thousand tonnes of CO ₂ eq					MtCO ₂ eq	MtCO ₂ eq	%
	2015	2018	2020	2025	2030	2018–30	2018–30	2018–30
LPG households, 1%								
Reference**	0.0	– 22.9	– 26.3	– 34.7	– 42.7	239.0	– 0.43	– 0.18
Extractive	0.0	– 23.4	– 26.9	– 35.6	– 43.5	371.4	– 0.44	– 0.12
LPG Transport, 1%								
Reference**	0.0	– 3.1	– 3.7	– 5.1	– 6.6	239.4	– 0.06	– 0.03
Extractive	0.0	– 3.2	– 3.7	– 5.3	– 7.0	371.8	– 0.07	– 0.02
Biomass efficiency, 1%								
Reference**	– 0.1	– 43.2	– 46.4	– 57.0	– 69.7	238.7	– 0.72	– 0.30
Extractive	0.0	– 45.0	– 48.6	– 59.5	– 72.6	371.1	– 0.75	– 0.20
Energy efficiency, 1%								
Reference**	– 0.1	– 1.1	– 1.4	– 2.9	– 4.8	239.4	– 0.04	– 0.01
Extractive	– 0.1	– 6.3	– 25.7	– 3.3	– 5.7	371.8	– 0.04	– 0.01

MtCO₂eq: Million tonnes of CO₂eq.

* Results are identical for all three Reference scenarios.

** Results are average of results across the three Reference scenarios.

recently updated emission reduction goal of 67.2 MtCO₂eq by 2030 as the country's Nationally Determined Contribution (NDC) to the UN Framework Convention on Climate Change, contains the first ever quantitative national emission target set by the Government of Mozambique (GoM). Our analysis shows that this emission reduction target is equivalent to a reduction of almost 30% in comparison to the basic Reference paths, and a 18% reduction in comparison to the Extractive development path. We show that this goal is very ambitious, unless Mozambique follows the Extractive development path that assumes exploitation of Mozambique's natural resources to its fullest potential. Furthermore, our policy analyses made it clear that in Mozambique sustainable energy policies at the supply-side lead to far higher cumulative emission reductions than demand-side policies. This is especially true for GHG emissions. Also, we showed that none of our supply-side energy policies – including a complete ban on thermal electricity generation capacity – would lead to shortages in the domestic electricity market, but it certainly would lead to a decrease in electricity export as compared to the no-policy scenarios, especially in the medium term. In terms of energy strategy, this supports the conclusion by van der Zwaan et al. (2018) that for Africa it is optimal to leapfrog CO₂-emitting technologies by massively deploying renewable options. Therefore, Mozambique indeed faces a trade-off between maximizing electricity export and maximizing emission reduction from

electricity generation.

Our analysis has shown that, despite the emergence of natural gas and coal, hydro will account for between 60% and 80% of total future electricity production. This hydro capacity is mostly generated from the Zambezi Basin, in the North of Mozambique. Several recent studies have assessed the vulnerability of major existing and planned new hydropower plants in the Zambezi River Basin (e.g. Spalding-Fecher et al., 2014; Spalding-Fecher et al., 2017; Stanzel and Kling, 2014; Yamba et al., 2011). They lead to the conclusion that the potential risk of climate change in this area is considerable, including both the risk of draughts that reduce run-off and reservoir storage capacity and the risk of growing frequency and severity of floods that may lead to shutdowns and damage of hydropower stations and the downstream transmission network. In addition, under a future drying climate it is likely that a water allocation priority issues emerges, with competition between water needed for irrigation versus water needed for electricity production. Hence, finding the optimal balance between renewable and thermal electricity generation in Mozambique—like in Africa at large—involves more than weighing the monetary benefits from electricity export against climate benefits from emission reduction. Amongst others, it implies that both climate change and upstream development must be explicitly incorporated into both project and system expansion planning.

In its most recent Africa Energy Outlook, the IEA (2014) concludes that the path from Africa's theoretical energy potential to harnessed modern sustainable energy supply for most of its population is likely to be long and complicated. Our study supports this conclusion by showing that policymakers in Mozambique face a major challenge in addressing the country's 'energy-dichotomy' by developing the large reserves of natural resources in such a way that it benefits its growing population and the non-extractive sectors of its economy. Nevertheless, our scenario analysis points out that Africa's energy future is neither deterministic nor gloomy: a country like Mozambique still has the option to avoid a lock-in into fossil fuels and benefit sustainably from its natural resource wealth and the rapid change and growth of the country's energy sector. The major challenge is to develop the appropriate institutional frameworks and economic incentives to do so.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.enpol.2018.10.056.

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